

Table for None

The Secret Science of Why We Can't Decide

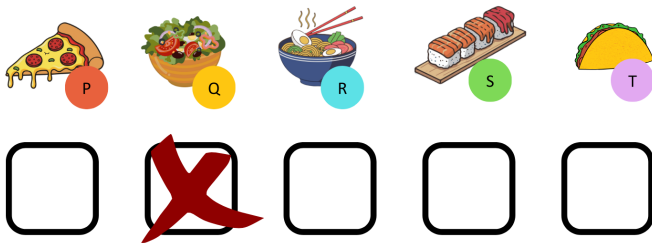
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Advisor: Professor de Silva

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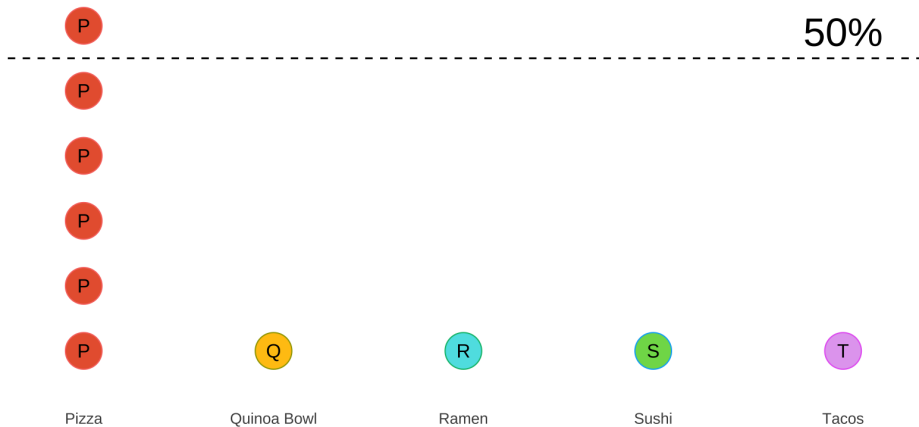




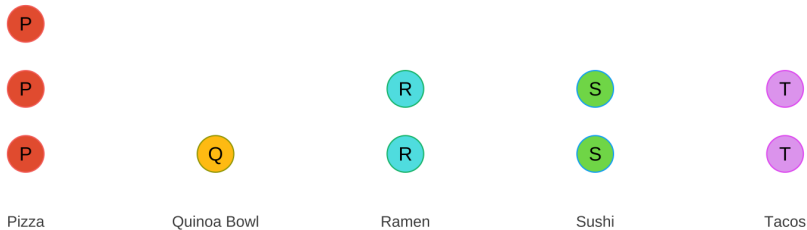




Simple Majority



Plurality



How many people are
actually HAPPY?

What Plurality Allows Us to See

	1st
P_1	P
P_2	P
P_3	P
P_4	R
P_5	R
P_6	S
P_7	S
P_8	T
P_9	T
P_{10}	Q



All of the Player's Preferences

	1st	2nd	3rd	4th	5th
P_1	P	R	Q	T	S
P_2	P	R	S	T	Q
P_3	P	R	T	S	Q
P_4	R	Q	S	T	P
P_5	R	Q	T	S	P
P_6	S	R	Q	T	P
P_7	S	R	Q	T	P
P_8	T	R	Q	S	P
P_9	T	R	S	Q	P
P_{10}	Q	R	T	S	P

Most Liked vs Least Liked

	1st	2nd	3rd	4th	5th
P_1	P	R	Q	T	S
P_2	P	R	S	T	Q
P_3	P	R	T	S	Q
P_4	R	Q	S	T	P
P_5	R	Q	T	S	P
P_6	S	R	Q	T	P
P_7	S	R	Q	T	P
P_8	T	R	Q	S	P
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P_{10}	Q	R	T	S	P

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P_7	S	R	Q	T	P
P_8	T	R	Q	S	P
P_9	T	R	S	Q	P
P_{10}	Q	R	T	S	P

Can we find a system taking all preferences into account, giving us
Ramen as the winner?

Choice	5 pts	4 pts	3 pts	2 pts	1 pts
Rank	1 st	2 nd	3 rd	4 th	5 th
P_1					
P_2					
P_3					
P_4					
P_5					
P_6	5	4	3	2	1
P_7					
P_8	T	R	Q	S	P
P_9	T	R	S	Q	P
P_{10}	Q	R	T	S	P

Borda Points in Our Van Scenario

	5 pts	4 pts	3 pts	2 pts	1 pts
P_1	P	R	Q	T	S
P_2	P	R	S	T	Q
P_3	P	R	T	S	Q
P_4	R	Q	S	T	P
P_5	R	Q	T	S	P
P_6	S	R	Q	T	P
P_7	S	R	Q	T	P
P_8	T	R	Q	S	P
P_9	T	R	S	Q	P
P_{10}	Q	R	T	S	P

	P	Q	R	S	T
P_1	5	3	4	1	2
P_2	5	1	3	1	2
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
P_{10}	1	5	4	2	3
Total	22	29	42	28	29

Ramen wins.

Does Borda Count have Any Issues?

Let's keep things simple.



P



Q



T

	1st	2nd	3rd
P_1	P	Q	T
P_2	P	Q	T
P_3	Q	T	P
P_4	Q	T	P
P_5	T	P	Q

	3 pts	2 pts	1 pts
P_1	P	Q	T
P_2	P	Q	T
P_3	Q	T	P
P_4	Q	T	P
P_5	T	P	Q

	P	Q	T
P_1	3	2	1
P_2	3	2	1
P_3	1	3	2
P_4	1	3	2
P_5	2	1	3
Total	10	11	9

We have a choice!

	3 pts	2 pts	1 pts
P_1	P	Q	T
P_2	P	Q	T
P_3	Q	T	P
P_4	Q	T	P
P_5	T	P	Q

Q wins



	3 pts	2 pts	1 pts
P_1	P	Q	T
P_2	P	Q	T
P_3	Q	T	P
P_4	Q	T	P
P_5	T	P	Q

CLOSED



THE TACO SHOP IS CLOSED

	3 pts	2 pts	1 pts
P_1	P	Q	X
P_2	P	Q	X
P_3	Q	X	P
P_4	Q	X	P
P_5	X	P	Q

CLOSED



THE TACO SHOP IS CLOSED

	2 pts	1 pts
P_1	P	Q
P_2	P	Q
P_3	Q	P
P_4	Q	P
P_5	P	Q

	2 pts	1 pts
P_1	P	Q
P_2	P	Q
P_3	Q	P
P_4	Q	P
P_5	P	Q

	P	Q	T
P_1	2	1	Closed
P_2	2	1	
P_3	1	2	
P_4	1	2	
P_5	2	1	
Total	8	7	

When we were considering **Pizza**, **Quinoa**, and **Tacos**...

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Quinoa

When we removed ~~Tacos~~ from the equation,

Pizza

Surely the outcome between **Pizza** and **Quinoa**, should only depend on those two?

Is there a better system out there?

- $\mathcal{A}$: set of alternatives

Pizza, Tacos, Quinoa ...

- V : set of voters

$P_1, P_2, \dots, P_n$

- $\mathcal{B}$: set of possible ranked ballots

$(P \succ T \succ Q), (T \succ P \succ Q), \dots$

- A voting system is a function, $F : \mathcal{B}^n \rightarrow \mathcal{B}_{\text{society}}$

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$$x \succ_i y$$

Person i prefers x over y

$$x \succ_S y$$

Society prefers x over y

The Trivial Cases: $|\mathcal{A}| < 2$

Case 0: No Alternatives ($|\mathcal{A}| = 0$)

If $\mathcal{A} = \emptyset$, the choice set is empty.

$$F(\mathcal{B}_{\emptyset}^n) = \emptyset$$

There is no decision to make. We don't make a choice.

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Case 1: The Singleton Set ($|\mathcal{A}| = 1$)

Let $\mathcal{A} = \{x\}$. Since every voter must rank x first (it is the only option):

$$F(\mathcal{B}_{\{x\}}^n) = x$$

Conclusion: Conflict is impossible.

Let $\mathcal{A} = \{x, y\}$.

Each voter i has strict preferences: either $x \succ_i y$ or $y \succ_i x$.

Proposed System: Simple Majority Rule

$$F(\mathcal{B}_{\{x,y\}}^n) = \begin{cases} x \succ_S y & \text{more people ranked } x \succ_i y \\ y \succ_S x & \text{otherwise} \end{cases}$$

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Odd Number of Voters

- **Simple Majority Rule** works perfectly.
- One alternative always receives $> \frac{n}{2}$ of votes.

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Even Number of Voters

- Risk a tie!
- Tie-Breaker: “Chairman” decides.

What do we even want out of F ?





Potential Desirable Characteristics

- **Anonymity:** it shouldn't matter who cast which vote. If two voters swap ballots, the outcome stays the same.
- **Strategy Proofness:** desirable system encourages you to vote for who you actually like.



Figure: Kenneth Arrow

1972 Nobel Prize

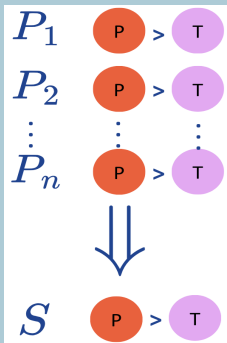
Changed the way we think about
democracy

Problem that had no solution for
close to 200 years

Unanimity

If all of society prefers x to y , then the social ranking must prefer x to y

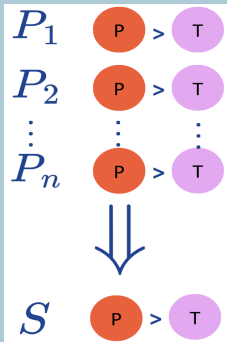
$$(\text{for all } i \in V, x \succ_i y) \implies x \succ_S y$$



Unanimity

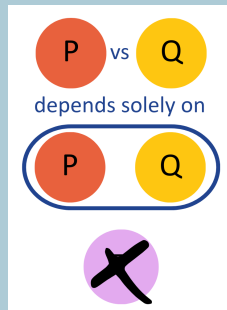
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Independence of Irrelevant Alternatives

The relative ranking of x and y given by our function depends only on the rankings of x with respect to y



Goal: It is really difficult to come up with a solution that we like.

Fix a valid voting system $F : \mathcal{B}^n \rightarrow \mathcal{B}_{\text{society}}$

Definition (Decisive Coalition)

A subset $A \subseteq V$ is **decisive** for the ordered pair (x, y) if:

$$(\text{for all } i \in A, \text{ suppose } x \succ_i y) \implies x \succ_{\mathbf{s}} y$$

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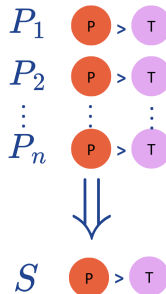
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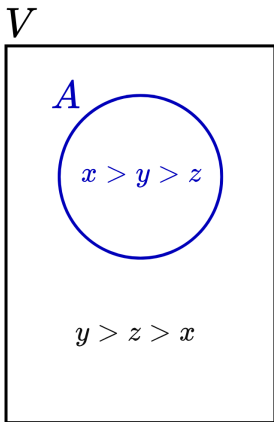
Lemma

We know there exists at least one set which is decisive by **unanimity**.

"If every voter in V preferred x to y , then society would have to prefer x to y ."



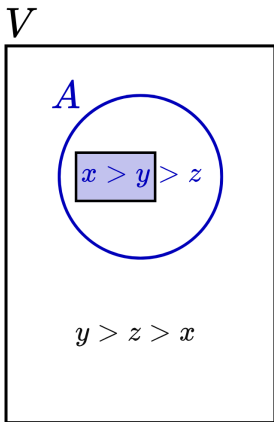
Given a subset $A \subseteq V$ which is decisive for (x, y) we will show group G is decisive for (x, z) .



A decisive over (x, y) by assumption

$$x >_s y$$

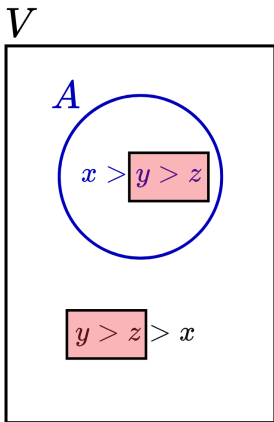
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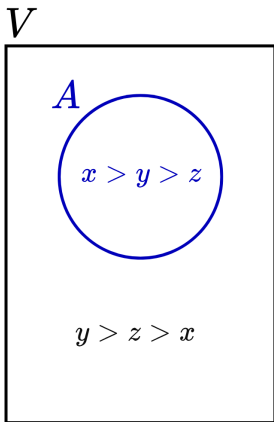
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V decisive over (y, z) by **unanimity**

$$y >_s z$$

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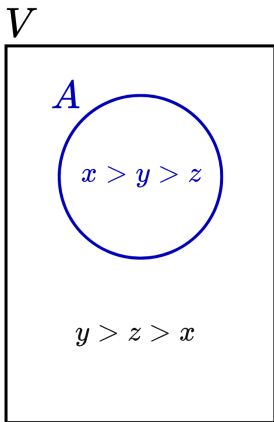
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via transitivity

$$x >_s y >_s z$$

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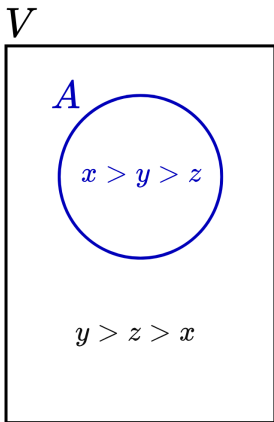


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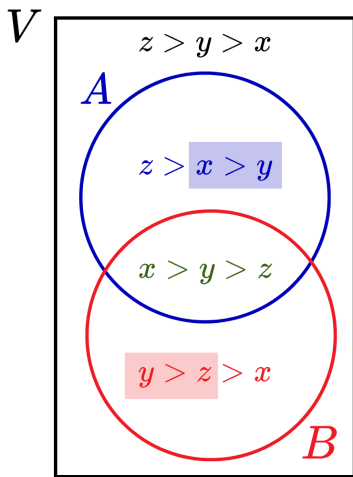
$$x >_s y >_s z$$

$$x >_s z$$

If there are at least 3 choices, we can show that if A is decisive for one pair, it must be decisive for all pairs.

What if we are not in agreement?

Suppose $A \subseteq V$ is decisive for (x, y) and $B \subseteq V$ is decisive for (y, z) , if power is “split”, it eventually must fuse to avoid contradictions.



A decisive over (x, y) by assumption

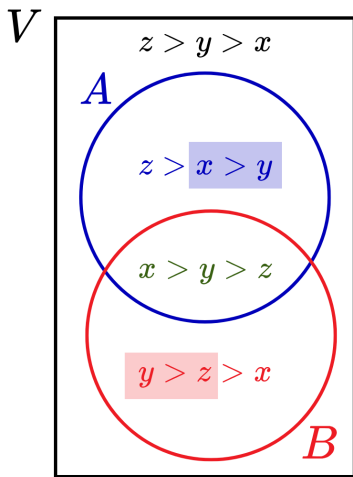
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B decisive over (y, z) by assumption

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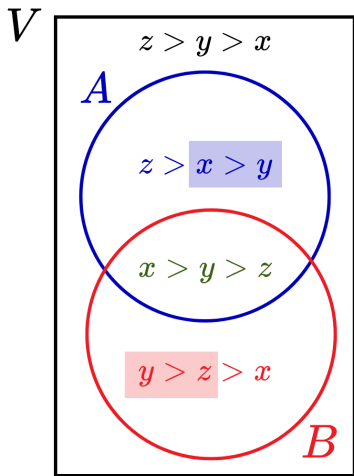
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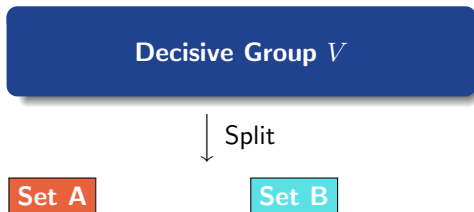
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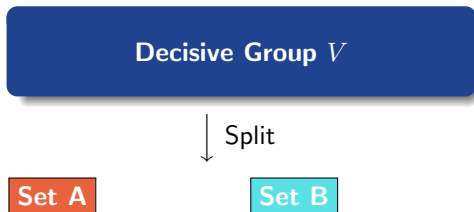
$$x >_s y >_s z$$

This set is sneakily powerful, contracting all of the power.

If we split a decisive group, power must fall to one side.



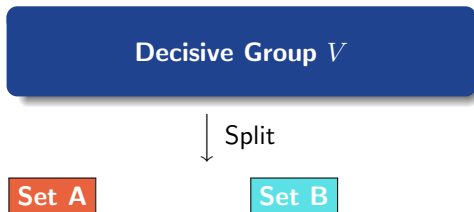
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The Scenario:

- **Set A** wants $x \succ y$
- **Set B** wants $y \succ x$

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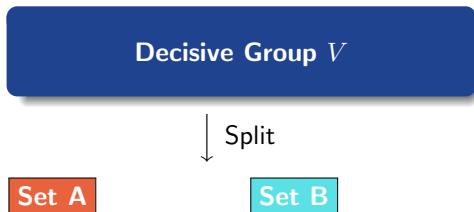
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The Result

To avoid a paradox, the system is forced to treat **either** Set A **or** Set B as the new Decisive Group.

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The Result

To avoid a paradox, the system is forced to treat **either** Set A **or** Set B as the new Decisive Group.

We can repeatedly “split” this power, coming to the realisation that fewer people are really calling the shots.

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The Dictator

An individual i such that if $x \succ_i y$, then $x \succ_S y$ *always*.

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Arrow's Theorem: The only system that satisfies all our rational assumptions... is a dictatorship.

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Thank you to my family and friends. A special shout out to all the professors in the Maths Department!

Thank you Professor de Silva!

